11. Sequences and Series

11.5. Alternating Series and Absolute Convergence

11.5 Learning Objectives
- Identify alternating series
- Determine if alternating series converge or diverge by the alternating series test
- Estimate the remainder for alternating series
- Distinguish between absolute and conditional convergence

Previously, we saw examples of series that were divergent when all of the terms of the sequence were positive, but the series became convergent when the terms had alternating signs.

11.5.2. Absolute Convergence

Definition 11.5.1

A series $\sum_{n=1}^{\infty} a_n$ is called **absolutely convergent** if the series of absolute values, $\sum_{n=1}^{\infty} |a_n|$ is convergent.

A series $\sum_{n=1}^{\infty} a_n$ is called **conditionally convergent** if the series is *convergent* but not *absolutely convergent*. (In other words, the series relies on the condition that some of its terms are negative to converge)

Theorem 11.5.1

If a series is absolutely convergent, then it is convergent.

Exercise 11.5.1

Prove this theorem. Hint: Use squeeze theorem.

Example 11.5.1

Determine whether the following series are absolutely convergent, conditionally convergent, or divergent.

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

Solution

The alternating harmonic series converges since $b_n = \frac{1}{n}$ goes to 0 by the alternating series test. However, the Harmonic series diverges because it is a p-series with $p = 1$. So, this series is conditionally convergent.

Example 11.5.2

Determine whether the following series are absolutely convergent, conditionally convergent, or divergent.

$$\sum_{n=1}^{\infty} \frac{\cos(n)}{n^3+2}$$

Solution

$$|a_n| = \left|\frac{\cos(n)}{n^3+2}\right| \leq \frac{1}{n^3+2} < \frac{1}{n^3}$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is a p-series with $p = 3 > 1$, it converges. So,

$$\sum_{n=1}^{\infty} \left|\frac{\cos(n)}{n^3+2}\right|$$

converges by direct comparison test. So

$$\sum_{n=1}^{\infty} \frac{\cos(n)}{n^3+2}$$

is absolutely convergent.

Example 11.5.3

Determine whether the following series are absolutely convergent, conditionally convergent, or divergent.

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3^n}{2^n+5^n}$$

Warning

$$\frac{3^n}{2^n+5^n} \neq \frac{3^n}{2^n} + \frac{3^n}{5^n}$$

This is not how fractions work!

Solution

For $n \geq 1$,

$$\begin{aligned} 2^n + 5^n &> 5^n \\ \frac{1}{2^n+5^n} &< \frac{1}{5^n} \\ \frac{3^n}{2^n+5^n} &< \frac{3^n}{5^n} = \left(\frac{3}{5}\right)^n \end{aligned}$$

and $\sum_{n=1}^{\infty} \left(\frac{3}{5}\right)^n$ is a geometric series with $r = \frac{3}{5} < 1$, it converges. So by the direct comparison test,

$$\sum_{n=1}^{\infty} \left| (-1)^{n+1} \frac{3^n}{2^n + 5^n} \right| = \sum_{n=1}^{\infty} \frac{3^n}{2^n + 5^n}$$

converges. So

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3^n}{2^n + 5^n}$$

is absolutely convergent.

11.5.3. Alternating Series Estimation

Again, we think about approximating the series by taking a partial sum s_n for some "large enough" value of n. If we do this for an alternating series, what can we say about the error in our estimation technique?

Theorem 11.5.2 Alternating Series Estimation Theorem

If $s = \sum_{n=1}^{\infty} (-1)^{n+1} b_n$ is the sum of an alternating series that satisfies:
(i) $b_{n+1} \leq b_n$ for all n
(ii) $\lim_{n \to \infty} b_n = 0$

then

$$|R_n| = |s - s_n| \leq b_{n+1}$$

Why does this work?

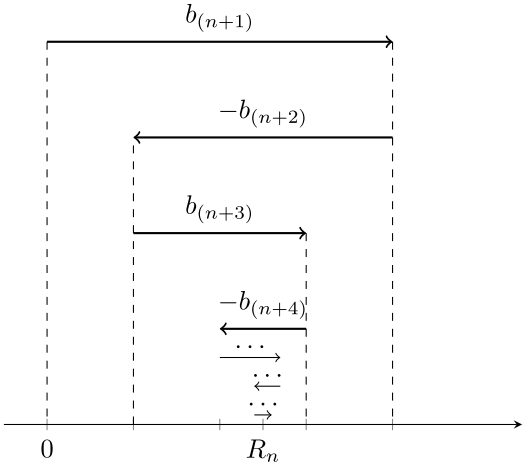
If

$$s = b_1 - b_2 + b_3 - b_4 + \ldots s_n = b_1 - b_2 + b_3 + \ldots \pm b_n$$

Then

$$|R_n| = |s - s_n| = \left| \mp b_{n+1} \pm b_{n+2} \mp b_{n+3} \pm \ldots \right|$$

Suppose it starts with $+b_{n+1}$ (but it works with $-$ also), then we can use a similar argument to why the alternating series test works.



The infinite sum for $|s - s_n|$ will have to be less than or equal to b_{n+1} since we are subtracting and adding smaller and smaller terms.

Example 11.5.4

Consider the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{2^{n-1}}$$

Estimate the alternating series with a partial sum, s_n, so that the absolute value of the error is less than $\frac{1}{50}$.

Solution

We want to find the n value where $|R_n| < \frac{1}{50}$. So, we need to find a term of the series where $|R_n| = |s - s_n| \leq b_{n+1} < \frac{1}{50}$. Then,

$$\begin{aligned} b_{n+1} &< \frac{1}{50} \\ \frac{1}{2^{n+1-1}} &< \frac{1}{50} \\ \frac{1}{2^n} &< \frac{1}{50} \\ 50 &< 2^n \end{aligned}$$

When $n = 5$, $2^n = 2^5 = 32! > 50$ and when $n = 6$, $2^n = 2^6 = 64 > 50$. So we need to use s_6 to estiamte our series.

$$s_6 = \frac{1}{1} - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} = 0.65625$$

and we know the error we get using 0.65625 as the value for the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{2^{n-1}}$ is no more than $\frac{1}{64} = 0.015625$.

11.5b Section Summary:

- We looked at different series and whether they are absolutely convergent, conditionally convergent, or divergent.
- We also discussed the error for using partial sums to approximate the value of alternating series.